FINAL PROJECT REPORT

Metro Ticket Generating System using ServiceNow

ServiceNow Global Certification Program

Project Name	Metro Ticket Generating System using ServiceNow
Platform	ServiceNow Now Platform
Core Components	Service Catalog, Flow Designer, Notifications, QR-code digital tickets

Project Summary

This project digitizes metro ticket booking through ServiceNow. Passengers select source and destination stations and enter travel details. The system calculates the fare automatically, processes the request through Flow Designer, generates a digital ticket with a QR code, and sends a booking confirmation using Notifications. The goal is to make ticket booking fast, accessible, and automated.

1. INTRODUCTION

1.1 Project Overview

Metro passengers need a convenient way to book tickets without depending entirely on manual ticket counters. A digital booking process can reduce waiting, simplify journey-detail selection, and provide immediate access to ticket information. This project proposes a Metro Ticket Generating System configured on ServiceNow, using a Service Catalog item as the booking form and automation to process each request.

The passenger selects a source station, destination station, and required travel details. The system validates input, calculates the fare using configured fare rules, and initiates ticket creation. A digital ticket containing a QR code is generated, and a notification confirms the booking. Flow Designer coordinates the automated workflow.

1.2 Purpose

The purpose is to provide a fast, accessible, and automated metro ticket booking experience, reduce manual effort, present fare information clearly, generate QR-code tickets quickly, and communicate booking status through notifications.

1.3 Objectives

• Provide a Service Catalog form for metro ticket booking.
• Capture source station, destination station, and travel details.
• Validate booking inputs and prevent incomplete submissions.
• Calculate the fare automatically using configured fare rules.
• Generate a digital ticket with a QR code after a valid booking.
• Use Flow Designer to automate ticket processing and Notifications to confirm the booking.

1.4 Scope

The core scope covers booking input, fare calculation, automated ticket creation, QR-code ticket generation, and booking notifications. Ticket retrieval, cancellation, refund processing, live train information, payment gateway integration, and QR validation at station gates are optional extensions and are not assumed to be implemented in the core version.

2. IDEATION PHASE

2.1 Problem Statement

Two problem statements describe the passenger's needs and operational requirements of the digital ticketing workflow:

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A metro passenger	Book a ticket quickly by choosing stations and travel details, and receive a usable digital ticket.	Manual booking may involve waiting, and passengers may be unsure of the fare or ticket status.	Booking, fare calculation, ticket generation, and confirmation may not be combined in one automated flow.	Impatient, inconvenienced, and uncertain whether the ticket is ready.
PS-2	A metro service operator or administrator	Process booking requests consistently and provide clear ticket information.	Manual processing can take time and ticket details may be communicated inconsistently.	A standardized catalog form and automated workflow may not be available in one process.	Concerned about delays, inconsistent data, and support requests.

2.2 Empathy Map Canvas

The empathy map considers the passenger's thoughts, feelings, observations, actions, pains, and gains while using a digital metro booking system.

THINK & FEEL	"I want to book quickly. Is the fare correct? Has my ticket been generated?"	**SEE**	ServiceNow booking form, station choices, fare information, QR ticket, and confirmation status.
HEAR	May hear travel advice, booking instructions, or information about metro ticket rules.	**SAY & DO**	Selects stations, enters travel details, reviews fare, submits booking, and opens the QR ticket.
PAINS	Confusing fields, incorrect station selection, unclear fare, delayed confirmation, or inaccessible QR code.	**GAINS**	Faster booking, automatic fare calculation, instant digital ticket, and clear notification.

2.3 Brainstorming and Prioritization

The main ideas to prioritize are the Service Catalog booking form, input validation, fare calculation, Flow Designer automation, QR-ticket generation, and confirmation notifications. Optional features such as cancellation/refund handling, dashboards, live service information, and gate-side QR verification can be considered after the core workflow is stable.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Passenger activity	System touchpoint	Goal / opportunity
Awareness	Needs to travel and looks for a convenient ticket option.	Metro travel need / portal entry point.	Make the booking option easy to find.
Enter	Opens portal and selects Metro Ticket Booking.	Service Catalog form.	Use clear labels and necessary fields.
Engage	Selects stations, enters travel details, and checks fare.	Station fields, travel details, fare calculation.	Validate choices and show fare before submission.
Exit	Submits request and waits for ticket.	Flow Designer, booking record, QR generation.	Show clear success or error status.
Extend	Receives confirmation and keeps QR ticket ready.	Notifications and digital ticket details.	Make QR clear and ticket details retrievable.

3.2 Functional Requirements

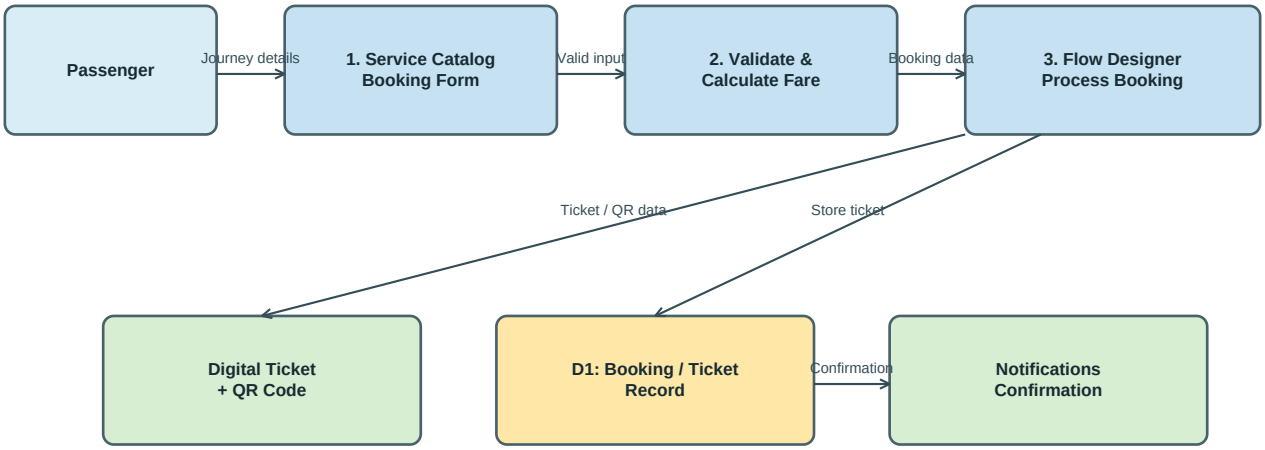
FR No.	Functional requirement	Sub-requirement
FR-1	Ticket Booking	Passenger opens a Metro Ticket Booking item through Service Catalog and submits journey details.
FR-2	Input Validation	Required fields are validated; source and destination should not be identical.
FR-3	Fare Calculation	Fare is calculated automatically using configured fare rules and displayed before submission where supported.
FR-4	Workflow Automation	Flow Designer processes a valid booking and creates the ticket/booking record.
FR-5	QR Ticket Generation	A digital ticket with QR code and unique ticket reference is generated after successful booking.
FR-6	Booking Notification	Notifications confirm ticket generation and provide ticket details or access instructions.

3.3 Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	The booking form should be clear, simple, and usable without extensive training.
NFR-2	Security	Only authorized users should access booking and ticket information.
NFR-3	Reliability	Booking submission and ticket creation should not lose valid request data.
NFR-4	Performance	The catalog form and automated workflow should respond promptly under normal load.
NFR-5	Availability	The booking service should be available subject to ServiceNow platform availability.
NFR-6	Scalability	The workflow should support additional stations and fare rules without major redesign.

3.4 Data Flow Diagram

The simplified Level-0 data flow below shows how booking information moves through the ServiceNow workflow, is processed and stored, and returns as a QR-code ticket and notification.



Flow: passenger input → validation/fare calculation → automated ticket creation → QR ticket and notification.

3.5 Technology Stack

Component	Purpose	Technology / ServiceNow feature
User interface	Passenger booking form	Service Catalog / Service Portal, depending on implementation
Workflow automation	Coordinate booking and ticket creation	Flow Designer
Fare processing	Apply configured fare rules	ServiceNow flow logic or scripts as configured
Ticket data	Store booking and ticket details	ServiceNow table / record
QR ticket	Create a QR representation of ticket data	QR-generation method configured for the project
Notifications	Send booking confirmation	ServiceNow Notifications

The exact QR-code generation method, table names, payment integration, and station-gate validation were not specified in the project description. Document these according to the actual configuration rather than assuming they are implemented.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The project addresses the passenger's need for a fast and accessible ticketing process. The proposed ServiceNow workflow combines journey-data capture, automatic fare calculation, ticket generation, and confirmation in a guided digital experience.

4.2 Proposed Solution

S. No.	Parameter	Description
1	Problem to be solved	Ticket booking can be time-consuming when journey details, fare confirmation, and ticket generation are handled through separate or manual steps.
2	Idea / Solution	Use Service Catalog to collect station and travel details, calculate fare, automate booking through Flow Designer, generate a QR-code ticket, and send a notification.
3	Novelty / Uniqueness	Combines booking input, fare processing, digital ticket output, and confirmation in one ServiceNow-based process.
4	Customer satisfaction	Passengers gain convenient access to tickets and fare information; automated processing can reduce repetitive manual work.
5	Business value	An internal digital service focused on convenience and operational efficiency.
6	Scalability	Station lists and fare rules can be extended as configuration permits; future features can be added in later iterations.

4.3 Solution Architecture

Passenger → Service Catalog booking form → validation and fare calculation → Flow Designer → booking/ticket record → QR-code digital ticket → Notifications → passenger. The exact technical implementation depends on the ServiceNow configuration.

4.4 Main Workflow

Step	Workflow activity
1	Passenger opens the Metro Ticket Booking catalog item.
2	Passenger selects source/destination stations and enters travel details.
3	System validates required fields and station choices.
4	Fare is calculated using configured fare rules.
5	Flow Designer processes the request and creates ticket information.
6	Digital QR-code ticket is generated and a confirmation notification is sent.

5. PROJECT PLANNING & SCHEDULING

5.1 Suggested Product Backlog

The following user stories are a proposed planning baseline. Sprint labels, story points, and priorities should be adjusted to match the actual project schedule and completed work.

Sprint	Epic	US #	User story	Points	Priority
Sprint 1	Booking	MTS-1	As a passenger, I can open the Metro Ticket Booking catalog item.	2	High
Sprint 1	Journey details	MTS-2	As a passenger, I can select stations and enter travel details.	3	High
Sprint 1	Validation	MTS-3	As a passenger, I receive clear validation for incomplete or invalid details.	2	High
Sprint 1	Fare calculation	MTS-4	As a passenger, I can see the fare calculated for my journey.	3	High
Sprint 2	Ticket generation	MTS-5	As a passenger, I receive a digital ticket with a QR code.	5	High
Sprint 2	Notifications	MTS-6	As a passenger, I receive confirmation after successful ticket creation.	2	High
Sprint 2	Ticket access	MTS-7	As a passenger, I can access ticket details when needed.	3	Medium
Sprint 2	Administration	MTS-8	As an administrator, I can review booking records for support.	2	Medium
Future	Enhancements	MTS-9	As a passenger, I may cancel or retrieve a ticket if those features are added.	5	Low

5.2 Suggested Sprint Plan

Sprint	Focus	Suggested duration	Exit criteria
Sprint 1	Catalog form, station fields, validation, fare calculation	1–2 weeks	Valid booking details can be submitted and fare is calculated.
Sprint 2	Flow Designer, ticket record, QR-code output, notifications	1–2 weeks	A valid booking creates a ticket and confirmation is sent.
Future	Ticket retrieval, cancellation/refunds, advanced validation	To be planned	Optional features are tested and accepted.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Test Cases

These proposed test cases verify the core workflow. Results should be marked only after execution in the actual ServiceNow instance.

Test ID	Scenario	Test steps	Expected result
FT-01	Required fields	Submit with required fields missing.	Submission is blocked and a clear validation message appears.
FT-02	Station selection	Choose different source and destination stations.	Selected stations are accepted.
FT-03	Invalid route	Choose the same source and destination.	System prevents submission or requests a valid destination.
FT-04	Fare calculation	Enter a supported station pair and travel details.	Fare follows the configured rules.
FT-05	Ticket generation	Submit a valid booking.	Ticket details and QR code are generated or made available as configured.
FT-06	Flow automation	Submit a valid request and inspect flow execution.	Flow Designer runs the configured workflow successfully.
FT-07	Notification	Complete a valid booking.	Confirmation is sent with ticket reference/details or access instructions.
PT-01	Response time	Time form load and submission under normal load.	Response meets the agreed performance target.
PT-02	Concurrent requests	Submit several bookings in a controlled test.	Requests are processed without unintended data loss.

6.2 Testing Status

No executed test results were supplied. This report lists test scenarios and expected results only; it does not claim that tests passed or that performance targets have been measured.

7. RESULTS

7.1 Expected Output

The intended output is a digital metro ticket booking workflow in which the passenger submits journey details through the Service Catalog, receives an automatically calculated fare, and obtains a QR-code ticket with booking confirmation. Add screenshots and execution evidence from the implemented ServiceNow instance.

7.2 Expected Benefits

• Faster and more accessible booking through a guided digital form.
• Reduced repetitive manual handling of booking details.
• Automatic fare calculation using configured journey rules.
• Digital QR-code ticket output for convenient access.
• Clear booking confirmation through Notifications.
• A reusable workflow that can be extended as requirements grow.

7.3 Limitations

The system depends on correct station data and fare rules. The QR-generation method and ticket validation process depend on the implemented configuration. Payment processing, cancellation/refunds, live train status, and integration with physical metro gates are not confirmed as core features.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

• Provides a structured digital booking process.
• Calculates fare automatically using configured rules.
• Generates QR-code tickets for convenient access.
• Uses Flow Designer to automate workflow steps.
• Uses Notifications to communicate booking confirmation.
• Built on ServiceNow and can be extended through configuration.

8.2 Disadvantages / Limitations

• Depends on availability and correct configuration of the ServiceNow instance.
• Incorrect station or fare configuration can produce incorrect details.
• QR-code generation and scanning require suitable implementation and testing.
• Passengers need access to the digital ticket or a supported retrieval method.
• Payment, refunds, gate integration, and live travel updates require additional scope.

9. CONCLUSION

The Metro Ticket Generating System proposes a ServiceNow-based way to digitize metro ticket booking. A passenger uses the Service Catalog to enter station and travel details; fare calculation and Flow Designer automate processing; the system generates a digital ticket with a QR code; and Notifications confirm the booking. The solution is intended to make ticket generation faster, more accessible, and more consistent. Claims about successful implementation should be supported by screenshots and tests from the actual ServiceNow instance.

10. FUTURE SCOPE

- Add ticket cancellation and refund workflows.
- Support ticket retrieval using a booking reference.
- Validate QR codes at station entry points through an appropriate integration.
- Integrate a payment gateway if online payment is required.
- Add live metro service information or route guidance.
- Add dashboards for booking volume, station demand, and workflow failures.
- Improve accessibility and mobile layout of the booking interface.
- Expand fare rules and station configuration as the metro network changes.

11. APPENDIX

11.1 Implementation Evidence

Add screenshots of the Service Catalog booking form, fare calculation, Flow Designer workflow, generated QR-code ticket, and confirmation notification. Use screenshots from your own ServiceNow instance.

11.2 Configuration Notes

Core components identified in the project description: Service Catalog, Flow Designer, and Notifications. Exact table names, scripts, QR-generation approach, payment integration, and deployed workflow configuration were not supplied and should be filled in from the actual project.

11.3 Project Summary

Project: Metro Ticket Generating System
Platform: ServiceNow
Core flow: Select stations and travel details → calculate fare → automate ticket generation → create QR-code digital ticket → send notification.